



# Effect of polymer nanoparticle morphology on fracture toughness enhancement of carbon fiber reinforced epoxy composites

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## ABSTRACT

Poor interlaminar fracture toughness has been a major issue in carbon fiber reinforced thermoset polymer composites. A number of techniques have been studied to improve this property, and resin matrix modification is the most direct and effective one. This study focuses on nanoparticle (NP) toughening of epoxy resins and how well the matrix toughness improvement transfers to its carbon fiber reinforced composites. The NPs used contain the same percentage of soft and hard polymer composition, but have different particle morphology, i.e. homogenous versus core/shell structures. Results show that the core/shell particles, especially when the soft polymer is the core and the rigid polymer is the shell, give 851%, 185%, and 43% enhancement in  $G_{IC}$  (epoxy),  $G_{IC}$  (composite) and  $G_{IIC}$  (composite), respectively, compared to the control specimen. Homogeneous particles are much less effective. Fracture surface morphology analysis suggests that effective toughening of composite requires the combination of well dispersed and strong particle-resin and resin matrix-fiber interfaces. This study provides insightful guidelines in selecting high-efficiency toughening particles, as well as comprehensive data on toughness transfer from neat resin to its corresponding composites.

## 1. Introduction

Carbon fiber reinforced polymer composites (CFRPs) are widely used to manufacture primary and secondary structures in aerospace, rail, automotive and energy industries due to their superior specific mechanical properties [1,2]. Most CFRP structures, either monolith laminates or sandwich panels, are built up by consolidating layers of unidirectional fibers, biaxial or multiaxial NCFs, or woven fabrics supported by the resin matrix, to maximize the in-plane mechanical performances. As a result, the lack of reinforcement in the through-thickness direction make CFRP susceptible to crack in the interleaf regions under transverse loading and result in delamination, especially upon impact [3,4]. Therefore, improving delamination resistance and interlaminar fracture toughness of CFRPs becomes a necessity.

Theoretical and experimental studies on delamination mechanisms in CFRPs have developed various methods to improve interlaminar fracture toughness, including through-thickness toughening [5–7], interlaminar toughening [8–11] and matrix toughening [12–16]. Because a matrix with high toughness can effectively absorb impact energy, resist damage and crack propagation, matrix toughening is the

most commonly used method to improve fracture toughness [17]. Furthermore, incorporating second-phase modifiers are considered as the most practical strategy for resin matrix toughening. Generally, three types of additive modifiers are often used, soft rubber prepolymers or their particles [18–20], soluble thermoplastics [21,22], and rigid organic or inorganic fillers [23–25]. It remains a challenge to improve the toughness without sacrificing the stiffness, or the other way around, when soft rubber or rigid inorganic particles are used as the toughener [26–28]. To address these issues, a method that combining both soft rubbery and rigid phase in the same formulation can be used. The soft rubbery phase is beneficial to the improvement of toughness while the rigid phase could compensate the reduction of stiffness. There have been different particle constructions of this nature, such as copolymers containing soft and rigid blocks, blends of soft and rigid particles, and core/shell structured particles, all have been proven effective in certain ways to toughen epoxy without significant decrease in mechanical properties. Chen et al. [29] used 12 wt% poly(methyl methacrylate)-b-poly(butyl acrylate)-b-poly(methyl methacrylate) (MAM) triblock copolymer in an epoxy-anhydride system, doubling the critical stress intensity factor ( $K_{IC}$ ) with slight reductions in the mechanical and thermal properties of the epoxy resin. Sprenger et al. [27]

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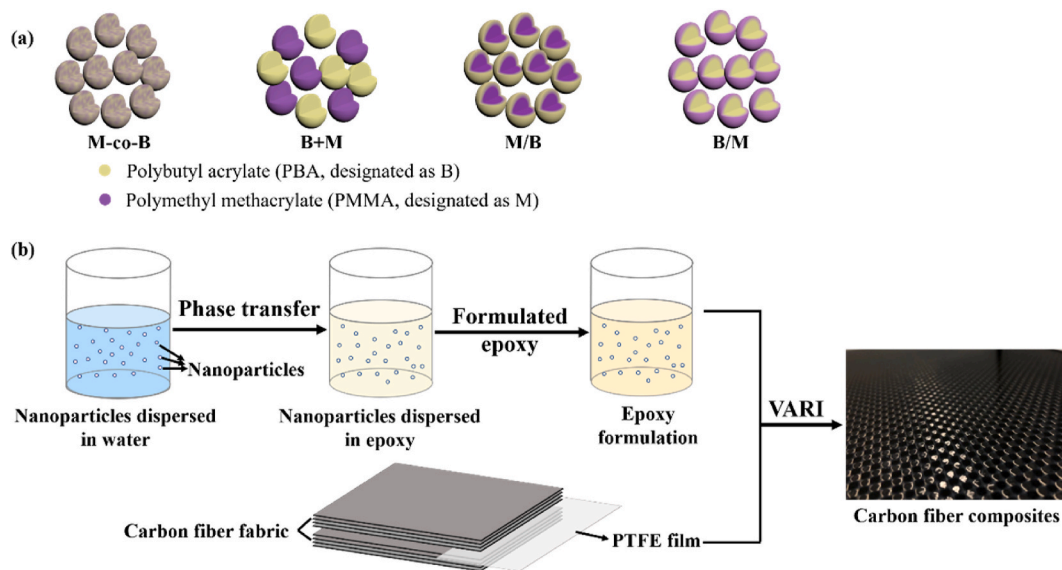


Fig. 1. (a) Schematic of the four polymer NP morphologies; (b) Illustration of particle phase transfer and composite laminate fabrication.

studied the toughening effect of an epoxy resin containing the mixture of soft rubber particles and rigid silica particles. They found that the fracture energy of mixed-particle-containing epoxy resin was greater than that of the individual particles, while the reduction in modulus was much less than that using individual soft rubber particles. Similar results were also demonstrated in our previous studies, that both the fracture toughness and tensile strength of core/shell particles were much greater than particles made of their corresponding homopolymers [30].

It is therefore evident that using soft and rigid NPs together can be effective than the individual ones when it comes to epoxy resin toughening. It is further demonstrated that combining the soft and rigid composition into a hybrid particle of particular heterogeneous morphology is even more effective. Admittedly, a number of studies are focused on constructing particles of different morphologies or structures and their effect on fracture toughness of the neat resins. Few systematical evaluations are conducted on fracture toughness of epoxy neat resins and their corresponding carbon fiber composites as a function of particle morphology when polymer compositions, particle size, loading level, resin content and fabrication process are kept the same. In addition, even fewer studies have been published about further investigation of the translation of such toughness enhancement to its corresponding carbon fibers reinforced composites.

Therefore, the overall objective of this work is to evaluate the influence of particle morphology on the fracture toughness of a carbon fiber reinforced composite, when the toughening effects of these particles are known for the neat epoxy resin used. In other words, this investigation aimed at exploring whether particle morphologies impart the same effects in both neat resin and its corresponding carbon fiber composites. To build on top of our earlier work [30], polybutyl acrylate (PBA, designated as B) was selected as the soft component and polymethyl methacrylate (PMMA, designated as M) as the rigid component. Four types of NPs having the same composition but different morphologies/structures, namely M-co-B copolymer particle, M + B particle blend, M/B core/shell particle and B/M core/shell particle, were prepared via emulsion polymerization. The size of these particles was carefully controlled at the nanometer level to be able to pass through the fibers during resin infusion, a vacuum assisted composite fabrication process used in this study. Double-cantilever-beam (DCB) Mode I and end-notched flexure (ENF) Mode II delamination tests were conducted to obtain composite interlaminar toughness. Tensile, flexural and interlaminar shear properties were also studied to help understand the particle effects. To reveal the toughening mechanisms, fracture surfaces

of the tested Mode I and Mode II samples were inspected by Scanning Electron Microscopy (SEM). The toughness transfer efficiency from the neat epoxy resin matrix to the carbon fiber composites was calculated and discussed.

## 2. Experimental section

### 2.1. Materials

The plain weave carbon fabric used has an areal density of  $480 \text{ g/m}^2$ , woven from SYT 45-12K carbon fiber (Zhongfu Shenying Carbon Fiber Co., Ltd., China.) An anhydride-cured epoxy was selected as the matrix resin due to its low viscosity and long open time, suitable for the resin infusion technique. The resin formulation consisted of the mixture of diglycidyl ether of bisphenol A (DGEBA from Sanmu Group Co, Ltd., China, EEW = 185) and neopentyl glycol diglycidyl ether (NGDGE from Huite Chemical, Ltd., China, EEW = 137) with a mass ratio of 60:40. Methyl hexahydrophthalic anhydride (MHHPA, J&K chemical, 98% assay) was used as the curing agent. 2-Ethyl-4-methylimidazole (2E4MI, Yuancheng Saichuang Technology Co., Ltd., China) was used as the catalyst.

Four polymer particles, as shown in Fig. 1a, were used as fracture toughness modifiers for the epoxy resins and its-based carbon fiber composites. The composition and synthesis procedure of these particles were reported in earlier publications [30]. These particles were phase-transferred and dispersed in mixture of DGEBA and NGDGE at a concentration of 36 wt%, also as described previously. The epoxy formulations containing the NPs were prepared by mixing the materials at room temperature. The stoichiometric ratio of the epoxy resin to the anhydride hardener was 1:1. For all samples, the loadings of polymer particles and catalyst (2E4MI) were kept at 10 wt% and 0.5 wt% based on total formulation, respectively. The formulated resin mixtures were degassed in a vacuum oven at  $60^\circ\text{C}$  for 1h before being infused.

### 2.2. Fabrication of the composite laminates

The composite laminates were fabricated via vacuum assisted resin infusion (VARI) process. 8 layers of  $32 \times 32 \text{ cm}$  carbon fabrics were arranged alone the warp direction ( $0^\circ$  orientation). A  $30 \mu\text{m}$  thick PTFE non-stick film was inserted at the laminate mid-plane (between ply #4 and #5) to act as crack initiator, as shown in Fig. 1b. Subsequently, the degassed epoxy resin was infused under vacuum at  $80^\circ\text{C}$ . After infusion,

**Table 1**

Fracture toughness and tensile properties of neat resin. (NP content is 10 wt%).

| Sample               | $K_{IC}$                |            | $G_{IC}$             |            | Tensile strength [MPa] | Tensile modulus [MPa] |
|----------------------|-------------------------|------------|----------------------|------------|------------------------|-----------------------|
|                      | [MPa m <sup>1/2</sup> ] | % increase | [J m <sup>-2</sup> ] | % increase |                        |                       |
| Control <sup>a</sup> | 0.71 ± 0.05             | 0          | 135 ± 18             | 0          | 66.9 ± 1.1             | 3289 ± 86             |
| M-co-B <sup>a</sup>  | 0.86 ± 0.02             | 21         | 242 ± 11             | 79         | 51.8 ± 3.1             | 2719 ± 10             |
| M + B                | 1.02 ± 0.01             | 44         | 329 ± 6              | 144        | 43.3 ± 3.0             | 2803 ± 50             |
| M/B <sup>a</sup>     | 1.18 ± 0.03             | 66         | 454 ± 20             | 237        | 42.6 ± 4.8             | 2681 ± 69             |
| B/M <sup>a</sup>     | 2.00 ± 0.05             | 182        | 1283 ± 66            | 852        | 57.1 ± 0.7             | 2791 ± 51             |

<sup>a</sup> Cited from Ref. [30].

the test panels were kept at this temperature for 30 min to promote fiber impregnation. Afterwards the panels were cured at 120 °C for 1 h, plus 140 °C for 2 h and 170 °C for 1 h. A control panel containing no particles was also prepared under the same condition.

### 2.3. Characterizations

Tensile strengths and fracture toughness of neat resin cast and CFRPs were measured to assess the effects of toughening particles. Dog-bone-shaped neat resin test coupons were prepared by pouring the resin into steel molds and cured as described above, their tensile strength and modulus were measured per ASTM D638. The fracture toughness,  $K_{IC}$  and  $G_{IC}$ , was measured using single-edge-notch bending geometry following ASTM D5045. The rheological behavior of uncured formulated resins containing NPs were tested on a TA DHR Rheometer with an angular frequency of 10 rad/s and a strain of 10%.

The interlaminar fracture toughness of the laminates were determined by Mode I Double Cantilever Beam (DCB) and Mode II End Notch

Flexure (ENF) tests according to ASTM D5528 and ASTM D7905, respectively. The Mode I and Mode II critical energy release rate,  $G_{IC}$  and  $G_{IIC}$ , were calculated as shown by Eq. (1) and Eq. (2).

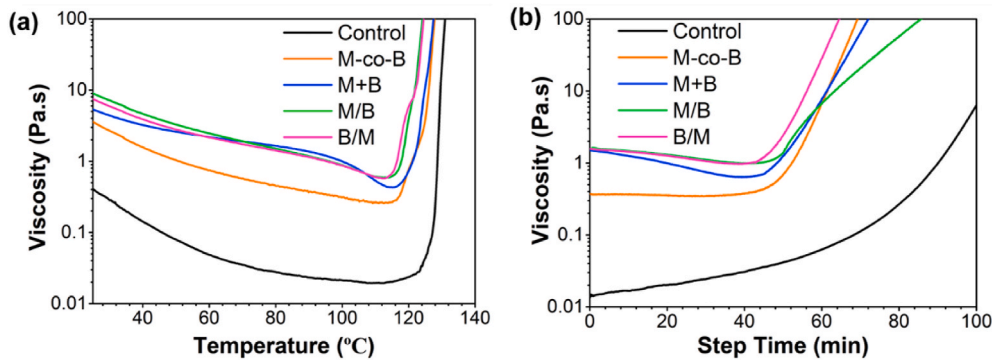
$$G_{IC} = \frac{3P\delta}{2ba} \quad (1)$$

Where P is the load (N),  $\delta$  is the displacement of the crosshead in the load point (mm), a is the delamination length (mm), b is the specimen width (mm).

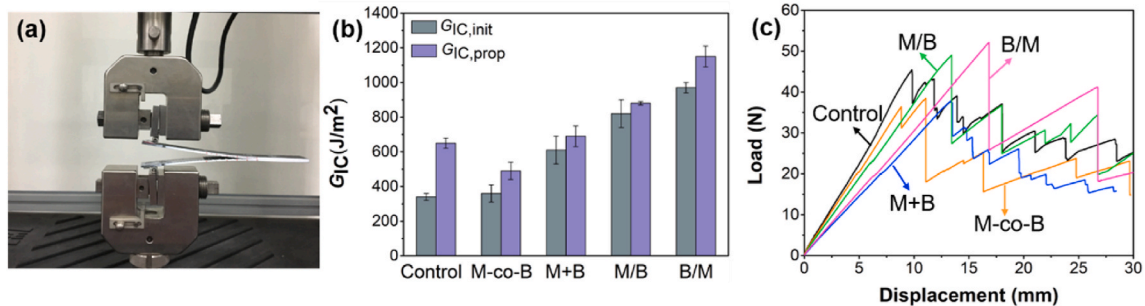
$$G_{IIC} = \frac{9a^2 P \delta}{2b(2L^3 + 3a^3)} \quad (2)$$

Where P,  $\delta$ , a and b are the same as in Eq. (1), 2L is the span length.

In the DCB test, piano hinges were attached to the top and bottom surfaces of the laminate using a structural adhesive. The side of the test specimens was marked with a white calibrated scale to better observe the crack propagation. During the test, the load was applied perpendicular to the crack plane with a crosshead speed of 1 mm/min. ENF specimens were tested using a three-point bending fixture and the mid-plane deflection was controlled by a crosshead speed of 1 mm/min. Interlaminar shear strength (ILSS) was determined by short beam shear (SBS) test per ASTM D2344. The tests were conducted using a span-to-thickness ratio of 4:1 with a crosshead speed of 1 mm/min. The flexural bending test was performed according to ASTM D790 with a crosshead speed of 1 mm/min, with a support span-to-depth ratio of 32:1, and 13 mm width. The above mechanical tests were conducted on a Wance ETM104B-EX electronic universal testing frame with a 2000 N capacity loadcell. The reported value for each type of the test was taken as the average of at least five valid measurements. The morphologies of the fracture surfaces from Mode I and Mode II interlaminar fracture toughness were observed on a Hitachi S4800 scanning electron microscope (SEM).

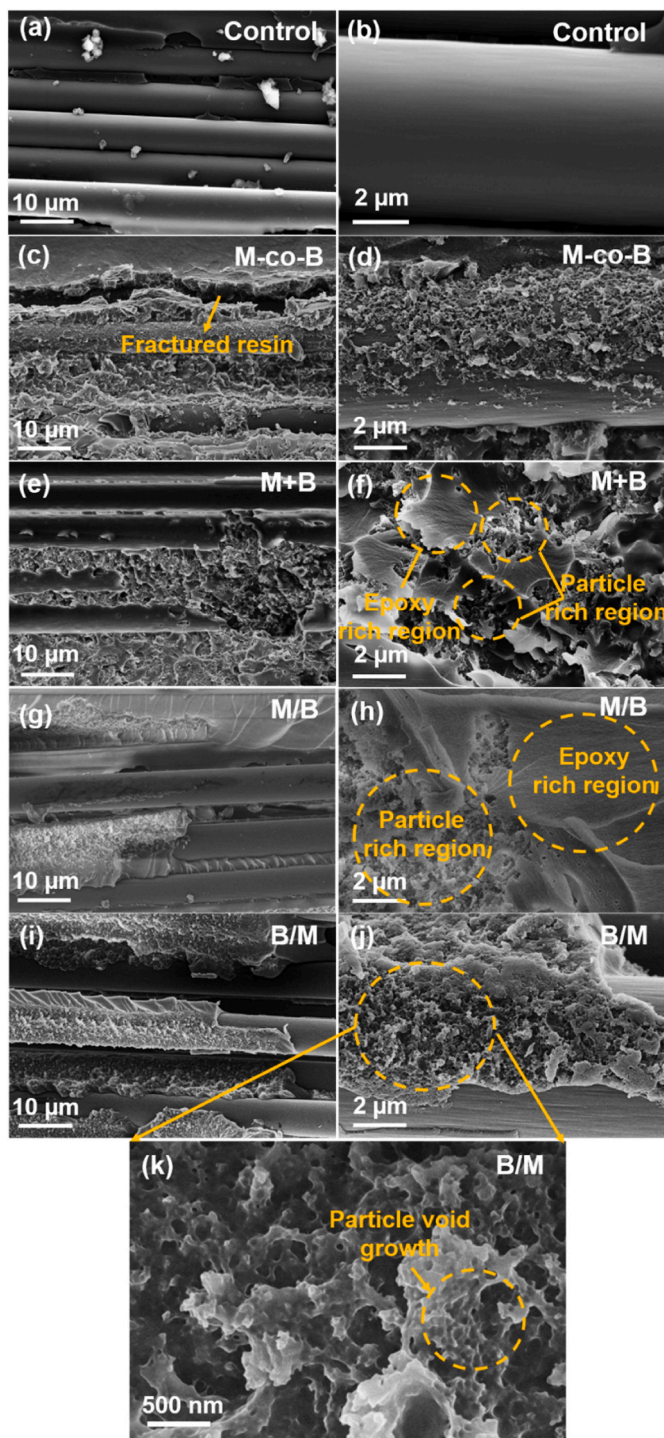


**Fig. 2.** (a) Viscosity as a function of temperature; (b) Time of the formulated resins at 80 °C. (NP content is 10 wt%).



**Fig. 3.** (a) Double cantilever beam test apparatus; (b)  $G_{IC,init}$  and  $G_{IC,prop}$  of control and 10 wt% NP-containing epoxy laminates; (c) representative load-displacement curves.





**Fig. 4.** SEM micrographs of DCB fracture surfaces of laminates under different magnifications: (a–b) Control; (c–d) M-co-B; (e–f) M + B; (g–h) M/B; (i–k) B/M. (NP content is 10 wt%).

### 3. Results and discussion

#### 3.1. Review of the neat resin properties

The NPs have been characterized and reported in earlier publication [30]. The particle size was carefully controlled to about 100 nm, with similar particle size distribution for all types. The core/shell morphologies of the NPs were revealed by transmission electron microscopy and thermal mechanical analysis. Particle loading was 10% by weight, as

previously found to provide significant toughening effect. Fracture toughness and tensile properties of the neat resins are presented in Table 1. As shown by data in this table, when compared to the control, in all cases adding NPs significantly increased stress intensity factor ( $K_{IC}$ ) and critical strain energy release rate ( $G_{IC}$ ). There were also evident differences between the particles in terms of toughening improvement, with M-co-B homogeneous copolymer particles to be the least effective, and B/M soft-core/hard-shell particles to be the most.

The differences in toughening effectiveness were resulted from the particle morphology and the degree of dispersion of the NPs in epoxy resin, because the polymer composition, NP size and size distribution were kept the same. Two conclusions can be further drawn from this data. First, the NPs with core/shell structure toughened better than those with a uniform morphology; second, NPs of soft-core/rigid-shell yielded higher fracture toughness than those of rigid-core/soft-shell. On the other hand, the addition of these NPs decreased tensile strengths. Although no tests were conducted to locate the root cause for such reduction, evidences from earlier study [31] suggested stress concentration effects caused by these particles, especially when the NPs form aggregations, were responsible for such behavior. The modulus reduction upon addition of the NPs was expected since the particles contained a soft polymer PBA, and its modulus was considerably lower than that of the epoxy resins.

It is previously established that when the outer layer of the particles is PBA, the particles tend to aggregate and are of weak interfacial interactions within epoxy matrix, and PMMA is opposite. Therefore, epoxy resins containing NPs M + B and M/B (having soft PBA in the outer layer) exhibited a large reduction in tensile strengths.

Despite the relatively low polarity of PBA in the outer layer of the NPs, which was unfavorable to the formation of strong interaction between the NPs and epoxy matrix, the presence of PBA was beneficiary to energy dissipation during crack propagation, which helped improve the fracture toughness. Therefore, as shown in Table 1, the NPs containing PBA (M + B, M/B and B/M particles) exhibited increases in fracture toughness. Again, M + B and M/B particles aggregated in epoxy resin, cutting down their toughening effectiveness, compared to B/M particles. Therefore, combination of effective energy dissipation and uniform dispersion is required for particle to be effective, and B/M particles precisely fit this model.

#### 3.2. Rheological behaviors of formulated resins

Rheological properties of the formulated resin are important for process control during composite fabrication by VARI which requires resin viscosity be sufficiently low at infusion temperature to wet and impregnate fibers, and eliminate porosity. The resin also needs to have reasonable open time to ensure consistent infusion, especially for large size parts. Addition of NPs normally increases resin viscosity significantly due to particle-particle and particle-resin interactions. Therefore, to determine an appropriate resin infusion temperature and time, the resin viscosity was measured as a function of temperature and time (see in Fig. 2). It can be seen from Fig. 2a that the control resin had the lowest viscosity, while the NP-containing resins exhibited much higher viscosities, as expected. Moreover, the viscosities of NP-containing resins showed stronger temperature dependence, owing to the breaking-out of interactions with increasing temperature. The sharp increase in viscosity denoted the gelation of the resins.

It is seen from Fig. 2a that viscosities were reasonably low for resins containing NPs above 80 °C. Because the higher the infusion temperature the shorter the resin's open time, and the onset temperature of curing was at 100 °C [30], the infusion temperature was selected to be 80 °C to balance the factors. Fig. 2b shows the isothermal viscosity at 80 °C, which showed more than 40 min open time for all resins, providing adequate infusion time.

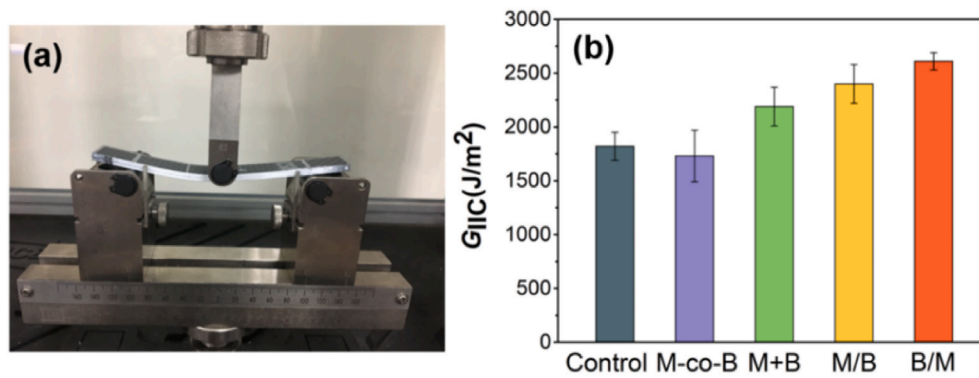


Fig. 5. (a) End-notched flexure test apparatus; (b) average  $G_{IIc}$  of control and 10 wt% NP-containing epoxy laminates.

### 3.3. Composite fracture toughness and mechanical properties

#### 3.3.1. Composite Mode I fracture toughness

Mode I fracture toughness using double cantilever beam test (DCB) is illustrated in Fig. 3a. Two values were defined, namely  $G_{IC,init}$  at which the load and displacement started to deviate from the straight line for the first loading, and  $G_{IC,prop}$  at which the load and displacement started to deviate from the straight line for the second loading [33]. The results were presented in Fig. 3b and Table S1 (Supporting Material). For the control specimen, the average  $G_{IC,init}$  and  $G_{IC,prop}$  were  $340 \text{ J m}^{-2}$  and  $650 \text{ J m}^{-2}$ , respectively. Comparatively, the incorporation of NPs, M-co-B, M + B, M/B and B/M, increased  $G_{IC,init}$  by 6%, 79%, 141%, 185%, respectively. The results were consistent with their neat resin fracture toughness properties, suggesting that the fracture energy at initiation were dominated by the resin toughness, which agreed with results reported by Shamsiah et al. [14].

The crack propagations energy,  $G_{IC,prop}$ , is usually a bit higher than its initiation fracture energies, due to the presence of fibers. Compared to the control sample,  $G_{IC,prop}$  showed increases of 35% and 77% with the addition of core/shell NPs, M/B and B/M, respectively. Contrarily, the uniform NPs, either M-co-B or M + B, did not help improving  $G_{IC,prop}$ . This effect is better seen from the load-displacement curves of the second loading in the DCB test, as shown in Fig. 3c, which showed decreasing crack loads for laminates with M-co-B and M + B particles, but laminates with M/B and B/M particles exhibited increasing crack loads. It is worth noting that these core/shell NPs had large load drops in the crack growth stages, which were characteristic for high toughness resins because of their high potential energy before crack initiation. When a large amount of potential energy releases, it will trigger large crack growth and result in large load drop.

The above DCB results showed that only NPs with core/shell structure improved fracture toughness for the carbon fiber reinforced composites, even though all the particles showed increased  $K_{IC}$  and  $G_{IC}$  compared with those of the neat resins. Furthermore, the NPs with a soft-core and hard-shell had higher toughness than those with hard-core and soft-shell.

To help understand the toughening mechanisms imparted by different NPs, fracture surfaces of DCB specimen were observed under SEM, shown in Fig. 4. The fracture surfaces of the control were clean and smooth with little resin attached to the fibers (Fig. 4a and b), displaying typical adhesive failure which was evident of poor fiber-matrix adhesion. When the NPs were added, cohesive failures were observed (Fig. 4c–j), indicating the increase in both resin toughness and adhesion to the fibers, which agreed with the results in Table 1. Consistent with the composite fracture results, the fracture surfaces of laminates containing uniform morphology particles M-co-B and M + B were very different from those of core/shell particles, M/B and B/M. Resins on the fracture surfaces of M-co-B and M + B (Fig. 4c and d and Fig. 4e and f), resulting from both adhesive and cohesive failures, did not exhibit the

integrity of a strong and resilient material, particularly for M-co-B which did not show improved toughness at all. In contrast, the core/shell NPs yielded fracture surfaces with plastic deformation, material cavitation and tearing (Fig. 4g and h and Fig. 4i and j), that would produce significantly higher forces and energy absorption. Once the cavitation had formed, it would relieve the triaxial stress ahead of the crack-tip, allowing the resin matrix to plastically deform and inducing plastic void growth, both would consume large amount of fracture energy and were responsible for the improved interlaminar fracture toughness [32, 33]. This effect was particularly visible for the highest performing B/M particles, as seen under high magnification (Fig. 4k).

#### 3.3.2. Composite Mode II fracture toughness

Mode II fracture toughness ( $G_{IIc}$ ) using end notch flexural test (ENF) (Fig. 5a) was conducted under three-point bending load, and only one cycle of load-displacement was recorded due to unstable crack propagation. Fig. 5b displays the average Mode II interlaminar fracture toughness ( $G_{IIc}$ ) for the control laminate and the NP-modified laminate, and the results are presented in Table S1 (Supporting Material). Compared with the control, the addition of M + B, M/B and B/M NPs into epoxy increased  $G_{IIc}$  by 20%, 32%, and 43%, respectively, whereas the addition of M-co-B slightly decreased  $G_{IIc}$ . Similar to the above  $G_{IC}$  findings, the core/shell NPs significantly outperformed their uniform-structure counterparts, and the soft-core/hard-shell NPs outperformed the hard-core/soft-shell ones.

Similar SEM observations were made on the ENF fracture surfaces as shown in Fig. 6. For the control laminates, clean fiber surfaces with little resin attached indicate poor fiber-matrix adhesion (Fig. 6a and b). In addition, shear hackle markings were evident in the interstitial matrix resin. The formation of shear hackle markings at  $45^\circ$  relative to crack propagation plane was attributed to tensile microcracks in the resin between adjacent carbon fibers [34,35]. The poor fiber-matrix adhesion was also observed for the laminate containing M-co-B NPs (Fig. 6c), suggesting that the NPs did not change the brittle nature of the resin (Fig. 6d). Closely resembled were the other uniform-structure NPs M + B (Fig. 6e). Not surprisingly, resins containing the core/shell M/B and B/M NPs adhered to the fibers without being stripped away by the delamination (Fig. 6g and i), indicating tortuous crack propagations. Crack deflections were also observed, as seen in the inset pictures Fig. 6g and i. Crack deflection was caused by off-plane force, thus generating new fracture surfaces and leading to increased strain energy required for crack propagation [36]. The SEM photographs (Fig. 6f, h and j) at high magnification again showed evidence of void formation. Voids resulting from particle cavitation and void growth are commonly accompanied with plastic deformation, yielding and crazing of the matrix resin, which are the main reasons for toughness enhancement [33,37].

It is noted that the above efforts to reveal the mechanisms of particle toughening is primitive. To fully understand why core/shell NPs toughen but the corresponding uniform NPs do not, and going one step



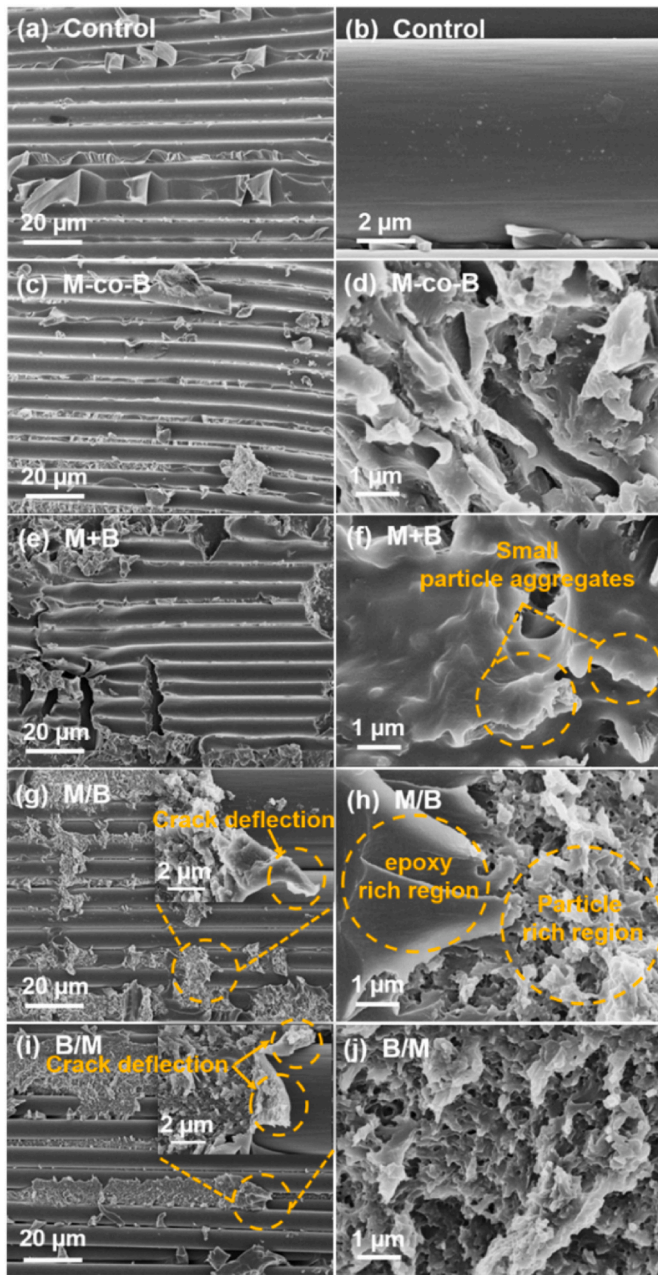


Fig. 6. SEM micrographs of ENF fracture surfaces of laminates under different magnifications: (a–b) Control; (c–d) M-co-B; (e–f) M + B; (g–h) M/B; (i–j) B/M. (NP content is 10 wt%).

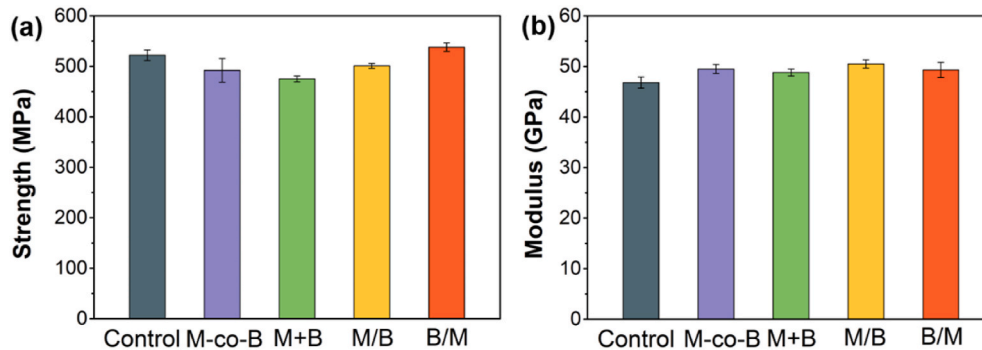


Fig. 7. Flexural properties of control and 10 wt% NP-containing epoxy laminates: (a) Flexural strength; (b) flexural modulus.

further, to understand why the soft-core/hard-shell NPs toughen better than the hard-core/soft-shell NPs, delicately designed experiments will have to be performed, together with sophisticated mathematical mechanical modeling and simulation, which would be a great subject for future research.

### 3.3.3. Laminate mechanical properties

In preceding sections, it was found that adding NPs lowered neat resin tensile strength and modulus. To see if this reduction in tensile properties is transferred to the corresponding carbon fiber reinforced

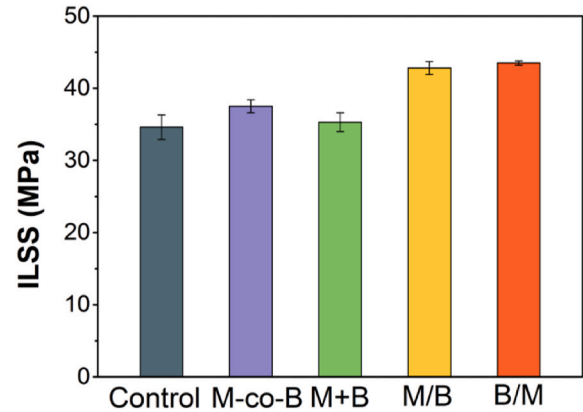


Fig. 8. Interlaminar shear strength (ILSS) of control and 10 wt% NP-containing epoxy laminates.

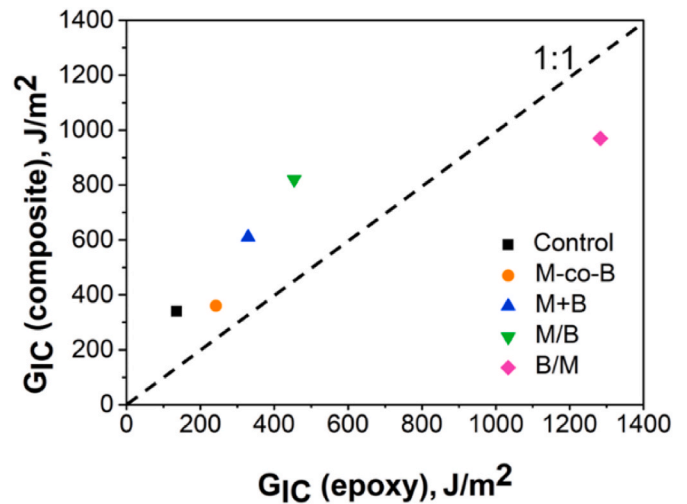


Fig. 9. Correlation of fracture toughness between neat resin and composite.

composites, flexural and interlaminar shear properties of the laminates with and without NPs were tested. All values were normalized according to laminate thickness and were listed in Table S2 (Supporting Material). It is shown that no significant reduction was observed for both flexural strength and modulus for all NP-containing epoxy laminates (Fig. 7). This might be due to that the flexural properties are fiber dominated, wherein the small difference in resin tensile properties would be insignificant.

The ILSS, which reflects the fiber-matrix bonding, were more in line with the fracture toughness, as shown in Fig. 8. The addition of core/shell NPs, M/B and B/M, increased ILSS by 24% and 26%, respectively, but the uniform-structure NPs, M-co-B and M + B, had little effect.

### 3.4. Correlating fracture toughness of neat resins and their composites

It is established that the fracture toughness of fiber reinforced polymer composites is determined by multiple factors, such as fracture toughness of matrix resin, laminate configuration, matrix-fiber interfacial bonding, presence of pores or defects and internal stress state [38]. Matrix resin fracture toughness is without doubt an important or sometimes a dominant factor. To show the toughness correlation of the neat resin to its carbon fiber laminates, their  $G_{IC}$  was plotted in Fig. 9. The diagonal line represents the 1:1 relationship between these two  $G_{IC}$  values, indicative of a perfect correlation or transfer. It can be seen from Fig. 9 that deviation, either positive or negative, was evident. Moreover, the observation that no perfect correlation occurred reflected the apparent interaction from the fibers. Nevertheless, the higher the neat resin  $G_{IC}$ , the higher its composite  $G_{IC}$ .

The correlation between resin  $G_{IC}$  (epoxy) and its corresponding composite  $G_{IC}$  (composite) has been studied by other researchers [39–42]. In general, the composite  $G_{IC}$  is higher than the corresponding resin  $G_{IC}$ , as in the cases of Control, M-co-B, M + B and M/B NPs (Fig. 9), due to the additional toughening contributed by the fibers. In addition, the observation that their toughness transfer from neat resin to composite lies above the diagonal line might be the evidence of fiber bridging, which significantly contributes to fracture energy increase, especially in cases of brittle resins. Indeed, as shown in Fig. 4, the fractured surfaces for B/M resin composite have somewhat more adhesive failure or interfacial failure than all other nanoparticle resin impregnated composites, probably indicating a little lowered interfacial bonding strength for the composite. While for high toughness resins, negative deviation from the diagonal line may result [14,41], as demonstrated by the resin and composite containing B/M core/shell NPs, which yielded the highest  $G_{IC}$ . As it was pointed out previously the toughness of a composite is not only determined by the resin alone but also by other factors. The major factors are the fiber, fiber/matrix interface and the resin/filler system [43,44]. In most cases, fiber properties are pretty much fixed when a particular fiber is selected as the reinforcement. Therefore, there are only two major factors left, namely fiber/matrix interface and the resin/filler interaction. In this study, we have focused on the modification of the resin/filler system by altering the morphology of the NPs. There is only so much that can be achieved by doing so since the crack propagation occurs only partially in the resin/filler phase while the fiber/matrix interface could also significantly contribute to the energy absorption in crack propagation during impact. This is confirmed by Fig. 9 where the improvement of composite  $G_{IC}$  as a function of resin  $G_{IC}$  has leveled off. It means that at this point, efforts should be made to improve fiber/matrix interface for further promotion of composite  $G_{IC}$  since we have achieved a high enough resin  $G_{IC}$  enhancement.

## 4. Conclusions

This paper studies the effect of NP composition and morphology on the fracture toughness of epoxy neat resins and their corresponding carbon fiber composites. Data showed that the effect of particle

morphology on composite flexural properties and interfacial shear strength was relatively small. However, the particle morphology had significant effect on the fracture toughness of epoxy neat resins and their carbon fiber composites. NPs having uniform morphology yielded little toughness improvement to the composites, while those with core/shell morphology were significantly more effective. Particularly, NPs with soft core/hard shell, i.e. B/M, produced the best toughening effect for both the neat resin and its composite. Specifically, the addition of 10 wt % of B/M particle gave 185%, 77% and 43% increase in  $G_{IC,init}$ ,  $G_{IC,prop}$  and  $G_{IIC}$  of carbon fiber composites, respectively, compared to the control. Scanning electron microscopy observation on the fracture surfaces demonstrated that the NPs altered fiber-resin interfacial adhesion, resulting in an improved interlaminar toughness. Furthermore, except for the B/M core/shell NPs, the composites with other NP-modified epoxy achieved a full transfer of epoxy matrix toughness to its carbon fiber laminate. This indicates that the toughness improvement due to modification of the NPs morphology has reached a plateau beyond which further enhancement of the composite  $G_{IC}$  may not be achieved by modifying the fillers alone.

## Author statement

**Na Ning:** Investigation, Data curation, Methodology, Writing-Original draft.

**Ming Wang:** Investigation, Testing.

**Gang Zhou:** Investigation, Testing.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compositesb.2022.109749>.

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